Gyuseok Lee

Ph.D. Student
School of Information Sciences
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Research Interests

Data Mining, Information Retrieval, Recommender System, LLMs

Education

- **Ph.D., School of Information Sciences, UIUC** — Aug 2025 - Present, Champaign, IL
  Overall GPA:
  Advisor: Prof. Dong Wang
- **M.S., Graduate School of Artificial Intelligence, POSTECH** — Feb 2022 - Feb 2024, Pohang, South Korea
  Overall GPA: 3.77 / 4.30
  Advisor: Prof. Hwanjo Yu
  Thesis: Collaborative Knowledge Distillation for Continual Learning in Recommender System
- **B.S., Computer Science, Mathematics & Statistics, Handong Global University (HGU)** — March 2015 - Feb 2022, Pohang, South Korea
  Overall GPA: 4.01 / 4.50 & Major GPA: 4.22 / 4.50 **(Magna Cum Laude)**
  Advisor: Prof. InJung Kim & Prof. Heonjoo Kim
  **Period includes 23 months of mandatory military service in the Republic of Korea Air Force.*

Research Experience

- **Visiting Researcher, Korea University** — April 2025 - Aug 2025, Seoul, South Korea
  Topic: Session-Based Recommendation with Validated and Enriched LLM Intents
  Advisor: Prof. SeongKu Kang
- **Researcher, POSTECH** — March 2024 - Aug 2025, Pohang, South Korea
  Topic: Continual Sequential Recommendation (Under Review at WSDM'26)
  Advisor: Prof. Hwanjo Yu
- **Visiting Researcher, University of Virginia (UVA)** — Sep 2023 - Feb 2024, Charlotteville, VA
  Topic: Collaborative Diffusion Model for Recommender System [WWW'25 short]
  Advisor: Prof. Jundong Li
- **Visiting Researcher, Carnegie Mellon University (CMU)** — Aug 2022 - Feb 2023, Pittsburgh, PA
  Coursework & Project: Deep Learning, Game Theory, NLP, Large-Scale Multimedia Analysis
  IITP AI Intensive Program, Sponsored by the Korean Government
- **Graduate Research Assistant, POSTECH** — Feb 2022 - Feb 2024, Pohang, South Korea
  Topic 1: Continual Collaborative Distillation for Recommender System [KDD'24]
  Topic 2: Calibrating Deep Learning Model for Medical AI Applications with Imbalanced Data [KICS'23]
- **Medical AI Research Intern, AITRICS** — Dec 2021 - Feb 2022, Seoul, South Korea
  Topic 1: Two-Stage Process with CNN and XGBoost for Thyroid Cancer Prediction
  Topic 2: Self-supervision for Classification of Pathology Images with Limited Annotations
- **Undergraduate Research Intern, HGU** — Feb 2021 - Dec 2021, Pohang, South Korea
  Topic: Development of Deep Neural Network Partial Learning Algorithm for On-Device Learning
  Advisor: Prof. InJung Kim

Publications

* indicates equal contribution.

1. **[SIGIR'26]** **Gyuseok Lee**, Yaokun Liu, Yifan Liu, Susik Yoon, Dong Wang, SeongKu Kang
   Session-Based Recommendation with Validated and Enriched LLM Intents (To be submitted)
2. **[WSDM'26]** **Gyuseok Lee**, Hyunsik Yoo, Junyoung Hwang, SeongKu Kang, Hwanjo Yu
   Capturing User Interests from Data Streams for Continual Sequential Recommendation **(Accepted)**
3. **[WWW'25 Short]** **Gyuseok Lee**, Yaochen Zhu, Hwanjo Yu, Yao Zhou, Jundong Li
   Collaborative Diffusion Model for Recommender System **(Accepted)**

Updated: Friday 24th October, 2025

4. **[KDD'24]** **Gyuseok Lee***, SeongKu Kang*, Wonbin Kweon, Hwanjo Yu
Continual Collaborative Distillation for Recommender System **(Accepted)**

5. **[KICS'23]** **Gyuseok Lee**, Hwanjo Yu
Calibrating Deep Learning Model for Medical AI Applications with Imbalanced Data **(Accepted)**

Research Projects

Recommender System

- **Session-Based Recommendation with Validated and Enriched LLM Intents** — April 2025 - Aug 2025, Korea University
This work leverages large language models (LLMs) to generate user intents from session data, aiming to capture users' hidden interests in a more interpretable way. The generated intents are seamlessly integrated into existing frameworks through mutual information maximization.
Role: First Author (To be submitted to SIGIR'26)

- **Capturing User Interests from Data Streams for Continual Sequential Recommendation** — March 2024 - Aug 2025, POSTECH
This work proposes a Transformer-based sequential recommendation model updated with non-stationary data streams. The goal of this model is to effectively preserve historical user interests while adapting to new user interests based on historical knowledge, enhancing the users' evolving preferences.
Role: **First Author to WSDM'26 (Accepted)**

- **Collaborative Diffusion Model for Recommender System** — Sep 2023 - Aug 2025, UVA
To mitigate the loss of personalized information during diffusion process when applying diffusion model to RS, we effectively leverage item-side information as auxiliary data.
Role: **First Author to WWW'25 Short (Accepted)**

- **Continual Collaborative Distillation for Recommender System** — March 2023 - Aug 2024, POSTECH
This work addressed the challenge of applying existing teacher-student knowledge distillation (KD) to a dynamic environment characterized by continuously incoming user-item interactions that are non-stationary and numerous. We deployed a compact model that not only preserves high performance through KD but also adapts to dynamic data, successfully integrating CL and KD for practical RS.
Role: **Co-first Author at KDD'24 (Accepted)**

- **Effective Visual Clustering for Personalized Multimodal Fashion Recommendation** — Sep 2022 - Dec 2022, CMU
This work integrated multimodal knowledge—including recommendation, text, and image data—to enhance fashion recommendations by efficiently handling large-scale data through clustering techniques.
Role: Project Leader - Led overall project progress from conceptualization to implementation.

Natural Language Processing

- **MLM is All you need** — Sep 2022 - Dec 2022, CMU
This project developed Question Answering (QA) and Question Generation (QG) models using a simple yet effective masked language model (MLM).
Role: Project Member - Problem Setting and Implementation.

- **Automatic Paper Assessment** — Sep 2022 - Dec 2022, CMU
This project developed a generative language model to assist the paper review process by extracting strengths and weaknesses and assigning an acceptance score based on the paper's content.
Role: Project Member - Problem Setting and implementation.

Computer Vision

- **Calibrating Deep Learning Model for Medical AI Applications with Imbalanced Data** — July 2022 - Jun 2023, POSTECH
This work calibrated a deep learning model for cancer prediction from chest X-rays using temperature scaling from Bayesian hyperparameter search. Focal loss was applied to handle class imbalance and further enhance calibration.
Role: **First Author to KICS'23 (Accepted)**

- **Self-Supervision for Classification of Pathology Images with Limited Annotations** — Dec 2021 - Feb 2022, AITRICS
This work applied self-supervision to predict cancer from pathology images with limited annotations. Six pretext tasks were designed to improve performance on the downstream task (i.e., cancer prediction).
Role: Intern – Problem Setting and Implementation.

- **Two-Stage Process with CNN and XGBoost for Thyroid Cancer Prediction** — Dec 2021 - Feb 2022, AITRICS
This work proposed a two stage approach for extremely large image classification in thyroid cancer prediction, combining CNN for feature extraction and XGBoost for classification. The results demonstrated the potential for practical applications in real-world medical AI services.
Role: Intern – Problem Setting and Implementation.

- **Developing DNN Partial Learning Algorithm for On-Device Learning** Feb 2021 - Dec 2021
  This work applied partial learning to enable efficient on-device learning for face image classification. HGU
  Role: Undergraduate Researcher – Problem Setting and Implementation.

Reinforcement Learning

- **Multi-Agent Reinforcement Learning for Taxi Repositioning with Attention Network** Sep 2022 - Dec 2022
  This project applied multi-agent reinforcement learning (MARL) to a ride-hailing system, aiming to optimize the balance between supply and demand through the use of attention networks. CMU
  Role: Project Leader - Led overall project progress from conceptualization to implementation.

Teaching Assistance

- **Artificial Intelligence** July 2025
  Assisted in implementing machine learning models using numpy, scikit-learn, and Pandas. SK (Company)
- **Artificial Intelligence** July 2024
  Assisted in implementing deep learning models (e.g., MLP, CNN, RNN, Transformer) using PyTorch. SK (Company)
- **Object Oriented Programming (OOP)** Feb 2023 - Jun 2023
  Assisted in teaching OOP concepts using C++. POSTECH
- **Big Data Analysis using R Programming** Sep 2021 Dec 2021
  Supported data analysis projects using R programming. HGU
- **Mathematical Statistics** July 2021
  Provided TA sessions on mathematical statistics. HGU
- **Statistical Methodology** Mar 2021 - Jun 2021
  Assisted in applying statistical techniques using R programming. HGU
- **Python Programming** Sep 2020 - Dec 2020
  Managed online discussion boards and provided feedback on coding assignments. KMOOC (Online Learning Platform)
- **Python Programming** Mar 2020 - Dec 2020
  Assisted in teaching basic Python syntax. HGU

Honors

- **Foundation Scholarship, GyeongWon Scholarship Foundation, South Korea** Nov 2021
- **Foundation Scholarship, GyeongWon Scholarship Foundation, South Korea** Mar 2021
- **Academic Excellence Scholarship, HGU, South Korea** Mar 2021
- **Academic Excellence Scholarship, HGU, South Korea** Sep 2020
- **Academic Excellence Scholarship, HGU, South Korea** Mar 2020

Skills

Python, PyTorch, TensorFlow, Linux, R programming, C++, C